

9. Ordinary generating functions: ordered arrangements, compositions, and recurrences

Combinatorics

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Suppose a staircase may be climbed using steps of several allowed heights. If we use exactly four steps, then the order of the four heights matters: the sequence $(1, 2, 1, 3)$ describes a different climb from $(3, 1, 2, 1)$. If the number of steps is not fixed, we must also account for every possible sequence length. Ordinary generating functions can record both the total height and the number of steps, while multiplication records the choices made in successive positions.

The same construction leads to compositions of an integer. Changing from an ordered list of parts to an unordered collection of parts leads instead to integer partitions and a different generating function. These examples also produce recurrences: a climb can be classified by its final step, and a recursive object can be classified by the smaller objects attached to its root.

After studying these notes, you will be able to do the following.

- Use the substitutions $x \mapsto x^d$ and $A(x) \mapsto x^r A(x^d)$ to impose divisibility and remainder conditions on exponents.
- Construct the OGF for ordered arrangements of exactly k objects by first assigning a separate variable to each position.
- Interpret the composition $A(B(x))$ as replacing each position of an outer arrangement by an inner object.
- Count climbs of a prescribed height with either a fixed or unrestricted number of allowed steps.
- Distinguish integer compositions from integer partitions and construct their generating functions.
- Derive Fibonacci and Catalan recurrences from combinatorial decompositions and translate those recurrences into OGF equations.

9.1 Substitution and restrictions on exponents

Let

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

Replacing x by x^d , where d is a positive integer, gives

$$A(x^d) = \sum_{n \geq 0} a_n (x^d)^n = \sum_{n \geq 0} a_n x^{dn}.$$

The coefficients have not changed, but the term that was in degree n is now in degree dn . Thus substitution multiplies every recorded size by d .

Proposition 1 (Scaling and shifting exponents). If $A(x) = \sum_{n \geq 0} a_n x^n$, $d \geq 1$, and $r \geq 0$, then

$$x^r A(x^d) = \sum_{n \geq 0} a_n x^{dn+r}.$$

Consequently,

$$[x^m]x^r A(x^d) = \begin{cases} a_{(m-r)/d}, & m \geq r \text{ and } m \equiv r \pmod{d}, \\ 0, & \text{otherwise.} \end{cases}$$

The geometric series

$$U(x) = 1 + x + x^2 + x^3 + \dots = \frac{1}{1-x}$$

records the choice of any nonnegative number of copies of one kind of object. Substituting x^2 for x gives

$$U(x^2) = 1 + x^2 + x^4 + x^6 + \dots = \frac{1}{1-x^2}.$$

Only even exponents occur, so this is the OGF for choosing an even number of copies. Multiplying by x shifts every exponent by one:

$$xU(x^2) = x + x^3 + x^5 + x^7 + \dots = \frac{x}{1-x^2}.$$

This is the OGF for choosing an odd number of copies. Notice that zero is included among the even choices. If a positive even number is required, the constant term must be removed:

$$\frac{x^2}{1-x^2} = x^2 + x^4 + x^6 + \dots$$

The same reasoning handles any remainder condition. For example,

$$\frac{x^2}{1-x^5} = x^2 + x^7 + x^{12} + x^{17} + \dots$$

records a number of copies congruent to 2 modulo 5.

Example 2 (Parity conditions in a product). A bag contains red, blue, and green tokens of each color in unlimited supply. How many selections of n tokens use an even number of red tokens, an odd number of blue tokens, and any number of green tokens?

The three color counts are independent and order does not matter. Their OGFs are, respectively,

$$\frac{1}{1-x^2}, \quad \frac{x}{1-x^2}, \quad \frac{1}{1-x}.$$

Therefore the required number is

$$[x^n] \frac{x}{(1-x^2)^2(1-x)}.$$

Each selected collection contributes once: its red, blue, and green counts choose one term from each of the three factors, and the exponents add to the total number of tokens.

9.2 A fixed number of positions in an ordered arrangement

Products have represented independent choices throughout this unit. For an ordered arrangement, the factors represent successive positions. Keeping a different variable for each position makes this interpretation visible before we combine the recorded sizes.

Suppose that one step may have a height in a set S of positive integers. For the moment, consider an arrangement of exactly three steps. Use u_1 for the first position, u_2 for the second, and u_3 for the third. The product

$$\left(\sum_{s \in S} u_1^s \right) \left(\sum_{s \in S} u_2^s \right) \left(\sum_{s \in S} u_3^s \right)$$

contains the monomial $u_1^{s_1}u_2^{s_2}u_3^{s_3}$ for the ordered sequence (s_1, s_2, s_3) . Because the variables retain the position numbers, the monomial $u_1u_2^2u_3^3$ is different from $u_1^3u_2^2u_3$.

To record only total height, substitute the same variable x for all three position variables:

$$u_1 = u_2 = u_3 = x.$$

The monomial for (s_1, s_2, s_3) becomes

$$x^{s_1}x^{s_2}x^{s_3} = x^{s_1+s_2+s_3}.$$

Different step sequences with the same total height now produce the same power of x , so their contributions add in its coefficient.

Let

$$B(x) = \sum_{s \in S} x^s.$$

After the substitution, the three-position product is $B(x)^3$. The same argument works for any fixed number of positions.

Theorem 3 (Ordered arrangements of fixed length). Suppose b_j types of objects have size j , and one object is placed in each position of an ordered arrangement. Let

$$B(x) = \sum_{j \geq 1} b_j x^j.$$

The number of ordered arrangements of exactly k objects and total size n is

$$[x^n]B(x)^k.$$

The assumption that the inner objects have positive size prevents a fixed total size from being obtained by inserting arbitrarily many size-zero objects.

Problem 4 (A fixed number of steps). A staircase allows steps of heights 1, 2, and 3. In how many ways can a climber reach height 7 using exactly four steps?

Solution: One position has OGF

$$B(x) = x + x^2 + x^3.$$

The four positions are ordered, so the requested number is

$$[x^7](x + x^2 + x^3)^4.$$

To extract this coefficient, write each step height as $1 + y_i$, where $0 \leq y_i \leq 2$. The four mandatory units account for height 4, leaving

$$y_1 + y_2 + y_3 + y_4 = 3.$$

Without the upper bounds, this equation has $C(6, 3) = 20$ nonnegative solutions. Exactly four of these have some $y_i = 3$, and no two variables can equal 3 when their sum is 3. Hence

$$[x^7](x + x^2 + x^3)^4 = 20 - 4 = 16.$$

Every solution gives one ordered four-step sequence, since $y_i + 1$ is the height of the step in position i .

If both total height and number of steps need to remain visible, introduce a second variable. Give every step an additional weight y . An arrangement of k steps then contributes y^k , and

$$1 + yB(x) + y^2B(x)^2 + y^3B(x)^3 + \cdots = \frac{1}{1 - yB(x)}.$$

Therefore

$$[x^n y^k] \frac{1}{1 - yB(x)} = [x^n] B(x)^k.$$

The exponent of x records total height, while the exponent of y records the number of steps.

9.3 Substitution as replacement of ordered positions

The preceding construction can be written as a substitution of one generating function into another. Let

$$A(t) = \sum_{k \geq 0} a_k t^k$$

count outer arrangements with k positions, and let

$$B(x) = \sum_{j \geq 1} b_j x^j$$

count the objects that may fill one position. The variables are deliberately different: t counts unfilled positions in the outer arrangement, while x counts the size contributed by a filling object.

An outer arrangement with exactly k ordered positions has weight t^k . Replacing each of those positions independently by a B -object changes that weight to

$$B(x)^k.$$

If there are a_k possible outer arrangements of length k , their total contribution is $a_k B(x)^k$. Summing over all k gives

$$A(B(x)) = \sum_{k \geq 0} a_k B(x)^k.$$

Thus the algebraic substitution $t = B(x)$ represents a combinatorial replacement: every outer position is filled by one inner object.

Proposition 5 (OGF substitution rule). Let $A(t) = \sum_{k \geq 0} a_k t^k$ count outer arrangements by their number of positions. Let $B(x) = \sum_{j \geq 1} b_j x^j$ count the allowed fillings of one position by size. If the positions of each outer arrangement are distinct, and each position is filled independently, then $A(B(x))$ counts the resulting objects by total size.

For an ordered sequence there is one outer arrangement of every length, so

$$A(t) = 1 + t + t^2 + t^3 + \dots = \frac{1}{1 - t}.$$

Substitution gives

$$A(B(x)) = 1 + B(x) + B(x)^2 + B(x)^3 + \dots = \frac{1}{1 - B(x)}.$$

The term $B(x)^k$ counts sequences of exactly k objects. The constant term 1 counts the empty sequence, which has length and total size zero.

Warning (A product of supplies counts a different set). For allowed step heights 2 and 5, the expression

$$\frac{1}{(1 - x^2)(1 - x^5)}$$

counts unordered selections of 2-steps and 5-steps. It records how many of each height are used, but it

does not record their order. Ordered step sequences are counted by

$$\frac{1}{1 - (x^2 + x^5)}.$$

These generating functions answer different questions.

Problem 6 (Allowed step lengths and any number of steps). A climber may use steps of height 2 or 5. How many ordered step sequences reach height 12?

Solution: One step has OGF $B(x) = x^2 + x^5$. A climb is an ordered sequence of such steps, so its OGF is

$$\frac{1}{1 - x^2 - x^5} = 1 + (x^2 + x^5) + (x^2 + x^5)^2 + \dots$$

We seek $[x^{12}]$. A sequence reaching height 12 can use six 2-steps and no 5-steps, which gives one sequence. It can instead use one 2-step and two 5-steps. The latter three steps may be ordered in 3 ways, according to the position of the 2-step. Therefore

$$[x^{12}] \frac{1}{1 - x^2 - x^5} = 1 + 3 = 4.$$

The unordered-selection product would give coefficient 2, one for each possible pair of multiplicities. It would not distinguish the three orders in the second case.

9.4 Compositions of an integer

Definition 7 (Composition). A **composition** of a positive integer n is an ordered list of positive integers whose sum is n . The entries in the list are called its **parts**.

For example, $(1, 3)$ and $(3, 1)$ are different compositions of 4. A positive part has OGF

$$P(x) = x + x^2 + x^3 + \dots = \frac{x}{1 - x}.$$

A composition with exactly k parts is an ordered arrangement of k positive parts, so its OGF is

$$P(x)^k = \left(\frac{x}{1 - x} \right)^k.$$

Theorem 8 (Compositions with a fixed number of parts). For $n \geq 1$ and $1 \leq k \leq n$, the number of compositions of n with exactly k parts is

$$[x^n] \left(\frac{x}{1 - x} \right)^k = C(n - 1, k - 1).$$

Proof. A composition $(a_1, \dots, a_k)$ of n has $a_i \geq 1$. Place n identical marks in a row. There are $n - 1$ gaps between consecutive marks, and choosing $k - 1$ of those gaps for separators produces k nonempty consecutive groups. Their sizes, read from left to right, are $(a_1, \dots, a_k)$. Conversely, the partial sums $a_1, a_1 + a_2, \dots, a_1 + \dots + a_{k-1}$ specify the separator gaps. The two constructions undo one another, so there are $C(n - 1, k - 1)$ compositions. $\square$

Allowing any positive number of parts gives

$$\begin{aligned} P(x) + P(x)^2 + P(x)^3 + \dots &= \frac{P(x)}{1 - P(x)} \\ &= \frac{x}{1 - 2x} \\ &= x + 2x^2 + 4x^3 + 8x^4 + \dots \end{aligned}$$

It follows that a positive integer n has 2^{n-1} compositions. This also follows from the separator description: each of the $n - 1$ gaps is either cut or not cut.

Example 9 (Compositions of 5 into three parts). The formula gives $C(4, 2) = 6$. The six compositions are

$$(1, 1, 3), (1, 2, 2), (1, 3, 1), (2, 1, 2), (2, 2, 1), (3, 1, 1).$$

Their different orders are retained.

9.5 Partitions of an integer

Definition 10 (Partition). A **partition** of a positive integer n is an unordered collection of positive integers whose sum is n . We write the parts in weakly decreasing order so that each partition has one representation.

The partitions $3 + 1$ and $1 + 3$ are therefore the same partition of 4, even though $(3, 1)$ and $(1, 3)$ are different compositions. To construct a partition, choose a nonnegative number of parts of size 1, a nonnegative number of parts of size 2, and so on. The choices for parts of size j have OGF

$$1 + x^j + x^{2j} + x^{3j} + \dots = \frac{1}{1 - x^j}.$$

The choices for different part sizes are independent, so their product is the partition generating function.

Theorem 11 (Euler's product for integer partitions). If $p(n)$ is the number of partitions of n , with $p(0) = 1$, then

$$\sum_{n \geq 0} p(n)x^n = \prod_{j \geq 1} \frac{1}{1 - x^j}.$$

For a fixed target n , factors with $j > n$ may be omitted.

Indeed, selecting x^{jm_j} from the j th factor means using m_j parts of size j . A term contributing to x^n is therefore a sequence of multiplicities $(m_1, m_2, \dots)$ satisfying

$$m_1 + 2m_2 + 3m_3 + \dots = n.$$

This multiplicity sequence determines one partition of n , and every partition gives one such sequence. For example, the seven partitions of 5 are

$$5, \quad 4 + 1, \quad 3 + 2, \quad 3 + 1 + 1, \quad 2 + 2 + 1, \quad 2 + 1 + 1 + 1, \quad 1 + 1 + 1 + 1 + 1.$$

Thus $p(5) = 7$, whereas 5 has $2^4 = 16$ compositions.

To record the number of parts as well as their sum, assign a factor y to each chosen part. The bivariate generating function is

$$\prod_{j \geq 1} \frac{1}{1 - yx^j}.$$

Its coefficient $[x^n y^k]$ counts partitions of n into exactly k parts.

9.6 Recurrences from the final step

A **recurrence relation** defines terms of a sequence using earlier terms. A recurrence must be accompanied by initial conditions, since the rule alone generally describes many sequences.

Let a_n be the number of ordered climbs of height n using step heights in a fixed set S of positive integers. Count the empty climb by $a_0 = 1$, and set $a_n = 0$ for $n < 0$. Every nonempty climb has a unique final step of some height $s \in S$. Removing that final step leaves a climb of height $n - s$. Conversely, appending an s -step to a climb of height $n - s$ produces a unique climb of height n whose final step has height s . The classes for distinct final-step heights are disjoint, so

$$a_n = \sum_{s \in S} a_{n-s} \quad (n \geq 1).$$

The OGF gives the same equation. If $B(x) = \sum_{s \in S} x^s$, then ordered climbs have OGF

$$A(x) = \frac{1}{1 - B(x)}.$$

Equivalently,

$$A(x) = 1 + B(x)A(x).$$

The term 1 represents the empty climb. In $B(x)A(x)$, the factor $B(x)$ chooses a distinguished final step and $A(x)$ chooses the shorter climb that precedes it. Comparing coefficients gives the recurrence above.

9.6.1 Fibonacci numbers from steps of heights one and two

For $S = \{1, 2\}$, let a_n count ordered climbs to height n . Classifying a nonempty climb by whether its final step has height 1 or 2 gives

$$a_n = a_{n-1} + a_{n-2},$$

with $a_0 = 1$, $a_1 = 1$, and $a_n = 0$ for $n < 0$. The sequence begins

$$1, 1, 2, 3, 5, 8, \dots$$

Define the Fibonacci numbers by

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad (n \geq 2).$$

Then $a_n = F_{n+1}$. The index shift occurs because the climb sequence starts with $a_0 = 1$, whereas the Fibonacci sequence starts with $F_0 = 0$.

The one-step OGF is $B(x) = x + x^2$, so

$$A(x) = \frac{1}{1 - x - x^2}.$$

Multiplying by the denominator gives

$$(1 - x - x^2)A(x) = 1.$$

For every $n \geq 2$, the coefficient of x^n on the left is $a_n - a_{n-1} - a_{n-2}$. Its vanishing recovers the Fibonacci recurrence.

9.6.2 Catalan numbers from a split into two smaller objects

The Catalan recurrence arises from a different decomposition. Let C_n count full rooted binary trees with n internal vertices, and let $C_0 = 1$ count the tree consisting of a single leaf. In a nontrivial tree, the root is an internal vertex with a left subtree and a right subtree. If the left subtree has i internal vertices, the right subtree must have $n - 1 - i$ internal vertices. The integer i may range from 0 to $n - 1$.

For a fixed i , there are C_i choices for the left subtree and C_{n-1-i} choices for the right subtree. The product rule and then the sum rule give

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i} \quad (n \geq 1).$$

Starting from $C_0 = 1$, this recurrence gives

$$C_1 = 1, \quad C_2 = 2, \quad C_3 = 5, \quad C_4 = 14.$$

Let

$$C(x) = \sum_{n \geq 0} C_n x^n.$$

The Cauchy product $C(x)^2$ records an ordered pair consisting of a left and a right subtree. Multiplication by x records the new internal root. Including the size-zero tree gives the OGF equation

$$C(x) = 1 + xC(x)^2.$$

The coefficient of x^n in $xC(x)^2$ is

$$\sum_{i=0}^{n-1} C_i C_{n-1-i},$$

so the generating-function equation and the Catalan recurrence express the same root decomposition. Solving this quadratic equation for $C(x)$ and extracting a closed formula are later steps; the recurrence already computes each term from earlier ones.

9.7 Ordered and unordered constructions compared

The form of a generating function depends on what constitutes one object.

- Exactly k ordered parts chosen from $B(x)$ are counted by $B(x)^k$.
- Any finite ordered sequence of such parts is counted by $1/(1 - B(x))$.
- Compositions use ordered positive parts, so their OGF is built from powers of $x/(1 - x)$.
- Partitions use a multiplicity for every possible part size, so their OGF is the product $\prod_{j \geq 1} (1 - x^j)^{-1}$.
- A recurrence results when every nonempty object has a unique decomposition into smaller objects, as in the final-step decomposition or the root decomposition of a binary tree.

Before selecting a formula, decide whether positions are ordered, what one exponent measures, and whether the number of parts is fixed. Those choices determine whether the construction requires a power, a geometric series of powers, or a product indexed by allowable part sizes.